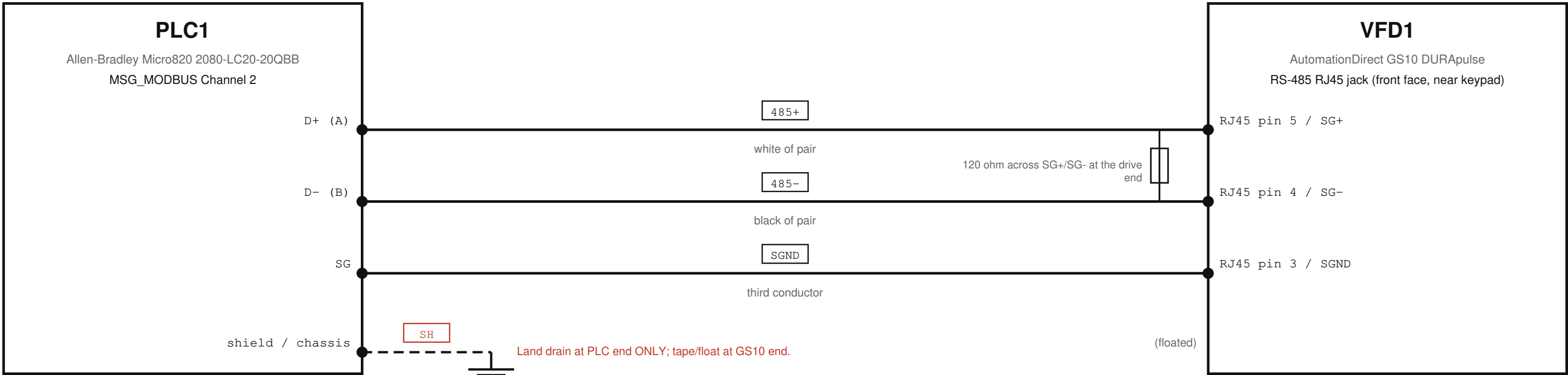


E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB (master) ↔ AutomationDirect GS10 DURApulse (node 1) · Modbus RTU only



CONNECTION TABLE

Src dev	Src terminal	Dst dev	Dst terminal/pin	Cable / conductor	Wire	Evidence	Notes
PLC1	D+ (A)	VFD1	RJ45 pin 5 / SG+	Belden 3105A pair · white of pair	485+	verified	Polarity: if silent/garbled, swap 485+/485- first.
PLC1	D- (B)	VFD1	RJ45 pin 4 / SG-	Belden 3105A pair · black of pair	485-	verified	
PLC1	SG	VFD1	RJ45 pin 3 / SGND	3rd conductor · third conductor	SGND	verified	Dedicated wire, NOT the shield.
PLC1	shield / chassis	VFD1	(floated)	tinned drain · drain	SH	field_verify	Land drain at PLC end ONLY; tape/float at GS10 end.

CCW SERIAL PORT: Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

Line params pending re-verify (OI-20): 2026-05-20 export read 38.4k/8N2; 2026-05-26 bench sniff read 9600/8N1 (shown). Fresh keypad/readback to adjudicate.

COMMAND WORDS (0x2000): STOP = 1 · FWD+RUN = 18 · REV+RUN = 34 · freq register 0x2001

34 NOT 20 — GS10_Integration_Guide.md (2026-03-16) and ControlsToVFD/original PhaseB.st print 20; corrected by PhaseB_V1.4 'FIX 2' (bit 5 REV + bit 1 Run), Beginner_Verify_V2 ('NOT 20'), WI-001 Table 9, and live Prog_init v2.1 (vfd_cmd_word := 34).

Beginner_Verify_V2 §4.4 (GS10 manual p.6); WI-001 Tables 8-10; PhaseB_V1.4.st; Prog_init_ConvSimple_v2.1.st

CORRECTED from May-16 draft: Channel 0 → 2 · SGND pin 1/8 → pin 3 · 8N2 → 8N1. Do not copy the old values.

NOTES

- Belden 3105A — shielded twisted pair (485+/485-) + 1 conductor (SGND) + drain
- VERIFIED (solid) = recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1; FIELD VERIFY (dashed) = exact PLC chassis ground point; cable P/N if not Belden 3105A.
- Port locations per CommsToVFD (PLC top-edge connector; GS10 front-face RJ45) — WI-001 says bottom-front. Wayfinding only; confirm on unit (FIELD VERIFY).
- GS10 RS-485 RJ45 cable confirmed landed (photo wire_4). The GS10 control connector (FWD/REV/DI/AI/AO1/DO1) is also wired — see OI-22.
- GS10 RS-485 RJ45 confirmed landed (full-res photo). The GS10 top run/DI control connector is EMPTY — no hardwired run/dir; command is Modbus. Bottom analog/output row is wired (OI-22).
- GS10 run/freq source = Modbus: P00.20=1, P00.21=2 (2026-05-20 export; same fact as E-006).
- P09.09 (Response Delay) = 10.0 ms — program flags CRITICAL (default 2.0 ms → ErrorID-55): Prog_init_ConvSimple_v2.1.st:79.
- 120 ohm termination (SG+/SG- at GS10 end): often omitted on a <2 m bench run — presence is FIELD VERIFY, not confirmed here.

SOURCES:

full-res photos review/photos/*_full.jpg + crops (2026-07-11); review/PHOTO_EVIDENCE_V5.md

DOC REFS:

CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

RS-485 / MODBUS RTU

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2

SHEET

E-007

REV A

2026-07-11

7 of 9